

Wrocław University of Science and Technology

Faculty of Management

Business Simulations

**Full report on the q-voter simulation
project**

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1 Project goals

1.1 General idea

This project proposes an innovative agent based Monte Carlo simulation to model the non-linear dynamics of customer churn by integrating social contagion theory with episodic external marketing stimuli. At the interaction layer, the simulation employs a modified non-linear q -voter model mapped onto a Watts-Strogatz complex network, where agents consult a group of q neighbors to determine their behavioral state, either conforming to a unanimous consensus or flipping their opinion independently when disagreement occurs. Instead of a uniform noise parameter or percolation grid, this framework introduces an internal cognitive variance for each agent using a heterogeneous baseline sensitivity, which mimics the individual structural breakdown of brand loyalty. The internal flipping probability is dynamically driven by the weighted influence of episodic advertising campaigns, which can either halve a loyal customer's susceptibility to peer pressure or boost a churned customer's likelihood to return, leading to a state transition that reflects a realistic push-and-pull between social consensus and targeted marketing noise.

1.2 What will the model be used for?

The primary objective of this model is to serve as a strategic risk assessment tool that identifies the critical thresholds where marketing interventions succeed or lose efficiency against the momentum of social contagion. By simulating various initial churn scenarios, the model will calculate the aggregate churn rate and map the long-term equilibrium concentration of loyal customers over time. This enables stakeholders to move beyond static churn metrics and instead conduct synthetic A/B testing on parameters like the number of influential neighbors consulted q , the boosted probability of returning due to successful ads, and the underlying randomness of the network topology unique to the Watts-Strogatz approach. From a technical perspective, the model functions as a scalable synthetic data generator driven by parallelized Monte Carlo iterations, revealing phase shifts where the system converges toward stable attractors of high loyalty or total churn. Ultimately, the model will provide a quantitative bridge between microscopic interaction rules and macroscopic system health, offering actionable insights into whether a customer base is approaching a "point of no return" dictated by the underlying structural dynamics of the network and episodic marketing bursts.

1.3 Why model this?

Modeling churn through this combined framework is essential because traditional predictive analytics often overlook the collective behavior and non-linear transitions inherent in complex social systems. By employing the modified q -voter model, we can observe how the system moves between mixed states of consumer opinion and ordered consensus phases depending on microscopic interaction rules, peer group sizes and network rewiring probabilities β . This approach allows the reveal of how episodic external incidents, such as targeted ad campaigns, act as recovery mechanisms that can either stabilize the system against attrition or fail to prevent a cascade into a high-churn state from which the brand cannot recover.

1.4 Who will be the agents?

The agents in this simulation are modeled as heterogeneous social actors, specifically customers within a brand ecosystem, who are characterized by a binary state variable representing either active engagement - loyalty or churn. Each agent's behavior is driven by two distinct mechanisms:

- an external social interaction layer following the rules of the non-linear q -voter model
- an internal individualized probability threshold reflecting their baseline susceptibility to influence.

On the macro scale, agents are embedded within a Watts-Strogatz complex network, where they periodically consult a random set of q neighbors to determine their next behavioral state. If the consulted neighbors exhibit a unanimous state of opinion, the agent conforms to that consensus. However, if there is a lack of consensus, the agent evaluates their own internal probability to act independently, which is heavily conditioned by whether they were recently impacted by an episodic ad campaign. This design ensures that agents are not merely homogeneous stochastic variables, but individualized entities whose final decision to churn or return is dictated by a dynamic interplay between local peer pressure, network topology, and the modifying shock of external marketing stimuli.

1.5 Tech stack and methodology

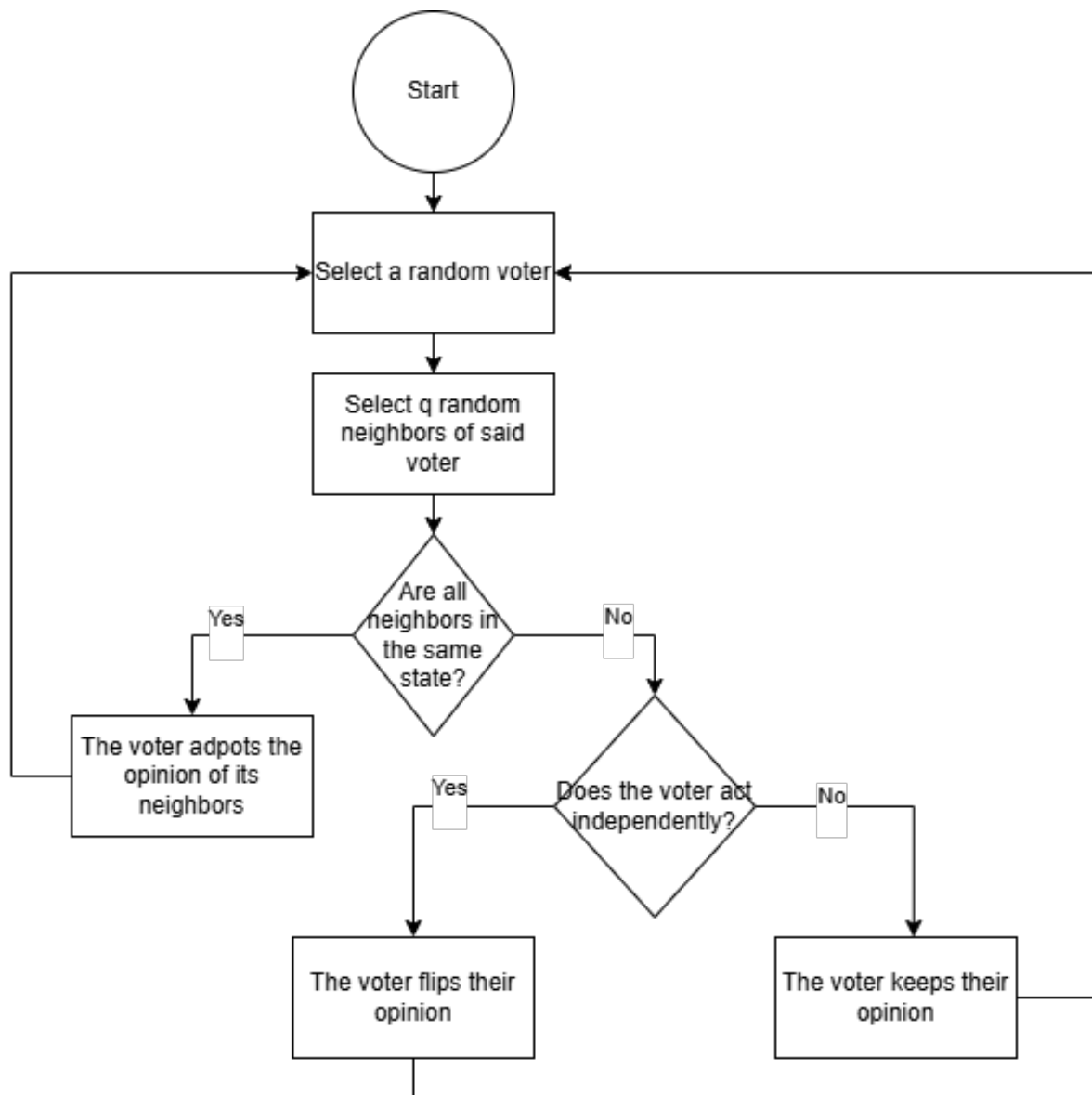
The project will be built using Python with NetworkX library to create social connections and NumPy to handle the high speed math of thousands of internal matrices. It will simulate the way opinions spread across different types of networks and tracks how long it takes for a group to reach a final decision.

1.6 Bibliography

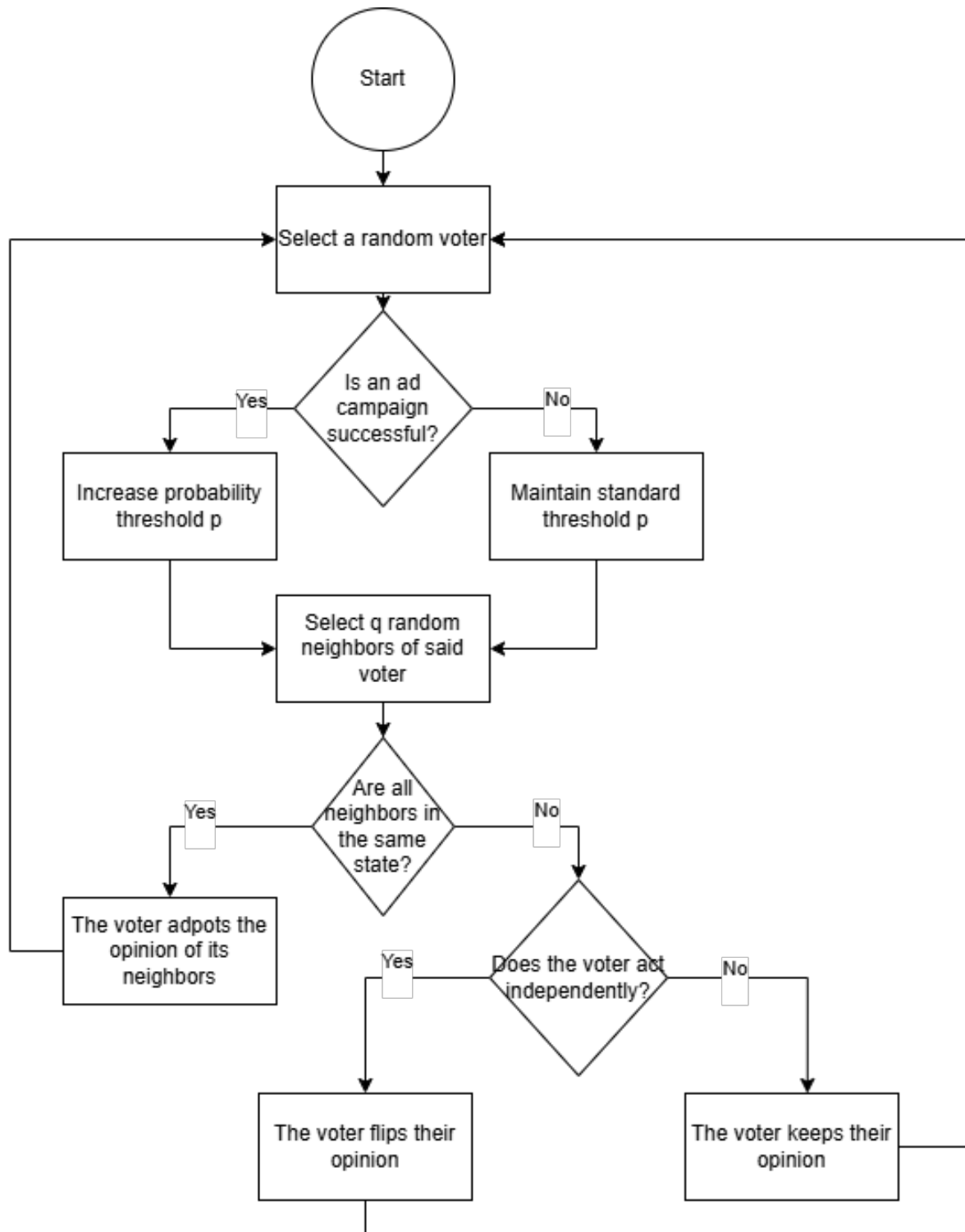
- <https://arxiv.org/abs/1206.1108>
- <https://arxiv.org/abs/0907.1775>
- **Research Gate article on threshold q -voter model**

2 Planned model flowcharts

2.1 Original model's flowchart



2.2 Modified model's flowchart



3 Program's interface and its features

My project is an ABM Monte Carlo simulation designed to model customer churn dynamics under the influence of episodic advertising campaigns. It combines the mechanics of the q-voter model with the complex network structures of a Watts-Strogatz graph. The program utilizes Streamlit to create an interactive online dashboard, divided into a configuration sidebar and a main viewing area for data visualization. The sidebar serves as the control center, allowing users to change the parameters of the simulation:

- Number of nodes $N \Rightarrow$ determines the total population of customers in the network, adjustable between 1000 and 100000
- k nearest neighbors $\Rightarrow$ sets the base connectivity of the Watts-Strogatz graph, controlling how many direct connections each node initially has
- q neighbours $\Rightarrow$ defines the size of the group an agent consults before deciding whether to change their opinion (as in conventional q-voter model)
- Boosted flipping probability $p_{inc} \Rightarrow$ represents the probability that a customer who has already churned will return after being exposed to a successful ad campaign.
- Monte Carlo steps $\Rightarrow$ control the duration of the simulation
- Tab 1 settings $\Rightarrow$ allow the user to set the starting churn rate and the total number of Monte Carlo simulations to run for establishing confidence intervals
- Tab 2 settings $\Rightarrow$ allow the user to specify how many runs should be averaged to plot the concentration lines for loyal customers.
- Network rewiring probability $\beta \Rightarrow$ accepts a comma separated list of values to test different network topologies, from a regular ring lattice of $\beta = 0$, to a highly random network $\beta = 1$

Upon clicking the "Run Simulations" button, the program compiles the Monte Carlo runs in parallel using Numba and generates two distinct visual tabs. First one displays a line chart plotting the mean customer churn rate over the simulation steps for each specified network rewiring probability value. It includes shaded regions which represent the confidence interval.

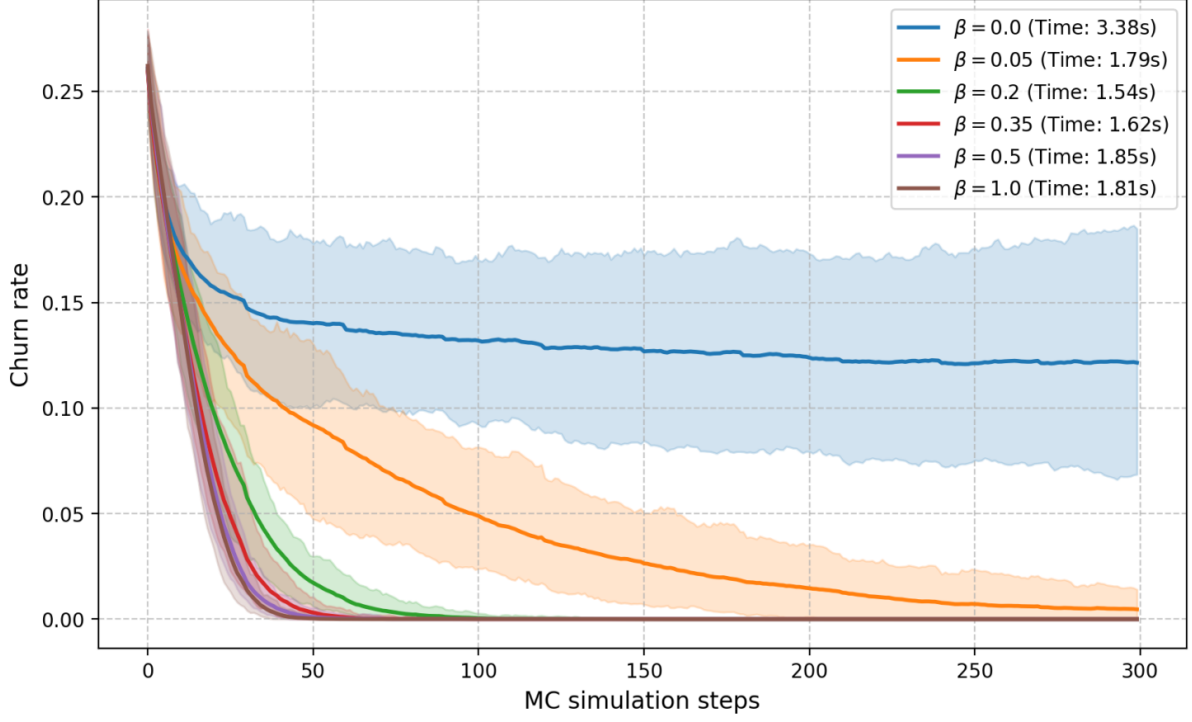


Figure 1: Churn rate per beta plot

The second tab generates a grid of charts, each one plotting multiple trajectories representing the concentration of loyal customers over time - each trajectory representing population of different initial churn values and each plot showcasing a different β value.

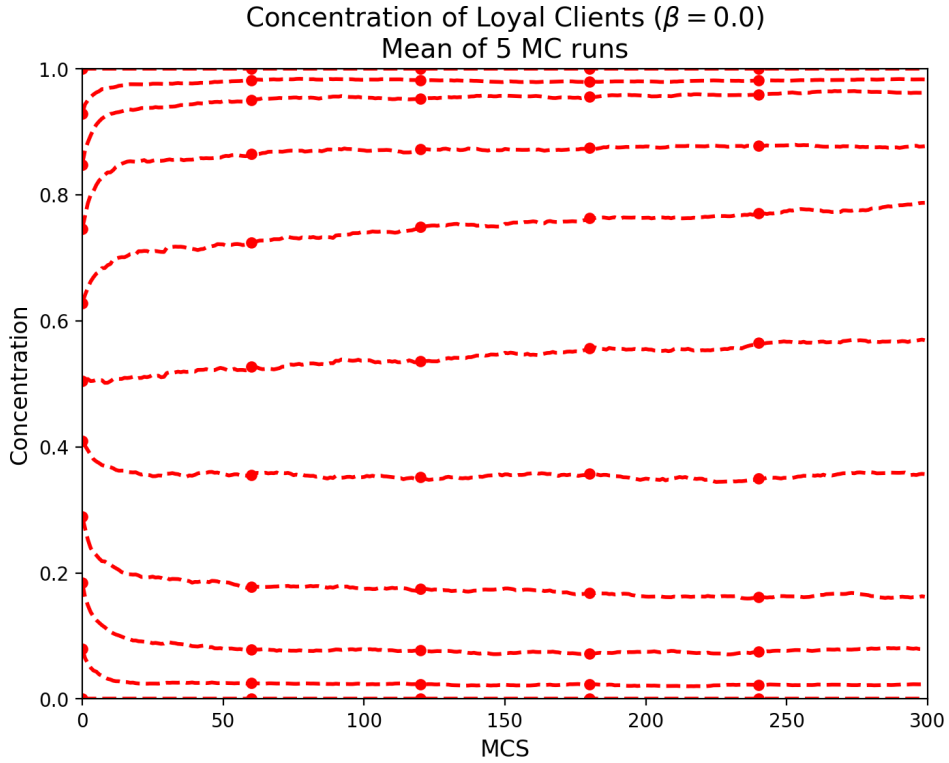


Figure 2: Loyal clients concentration example

4 Simulation results

The simulation is driven by a mix of peer pressure and external marketing influences:

- Ads are deployed episodically, specifically every 10% of the total Monte Carlo steps
- Based on cited marketing literature, each ad randomly impacts between 5% and 20% of the customer base
- If a loyal customer sees an ad, their personal probability to churn is cut in half, making them more resilient to peer pressure
- If a churned customer sees an ad, they use the boosted p_{inc} probability to decide if they will return
- If a customer is not currently impacted by an ad, they rely on their own baseline sensitivity to peer pressure
- During a step, an agent looks at q random neighbors. If all neighbors share the same opinion, the agent conforms to it. If the neighbors disagree, the agent flips their opinion based on their current probability threshold

Based on the underlying logic and execution outputs, the simulation yields several insights. As one could imagine, the overall churn rate does not remain static, rather dynamically shifts from the initial state defined by user, as peer influence and episodic ads take effect. Moreover, the computational outputs show that calculating results for different rewiring probability β takes roughly 47 to 57 seconds per 100 runs. As β value increases, the network tends to polarize more quickly. Lastly, by plotting concentrations from 0 to 100% initial loyalty, the simulation reveals the system's "attractors" or steady states. Regardless of whether the company starts with a completely loyal base or a completely churned base, the combination of the q-voter peer pressure and the p_{inc} ad boosts will drive the overall customer concentration toward specific equilibrium points over the course of the Monte Carlo steps.

5 Sensitivity analysis

5.1 Macroscopic observables characterizing the simulation's dynamics

To fully grasp the dynamics of the simulated customer base, the most effective macroscopic quantities to observe are definitely the aggregate churn rate (which is the total fraction of the network which has churned at any given Monte Carlo step) and the stationary concentration of loyal customers (results of the plots showcased in the second tab on Streamlit app. It defines the system's long term equilibrium and reveals the "attractors", identifying to which the customer base naturally converges).

5.2 Parameter ranges worth investigating

To capture the full spectrum of possible behaviors, specific ranges are particularly valuable to explore:

- Network rewiring probability $\beta \Rightarrow$ the range from 0.0 (a perfectly ordered ring lattice) to 1.0 (a completely random graph) is critical. Investigating the 0.01 to 0.1 range is highly recommended, as this is the classic "small world" regime where network dynamics undergo drastic shifts
- Initial loyal fractions $c_0 \Rightarrow$ scanning the full spectrum from 0.0 to 1.0 (by steps of 0.1 value) is essential to find tipping points where the majority opinion flips
- Conformity group size $q \Rightarrow$ testing $q \in [2, 6]$ range mimics realistic human peer groups. Higher values of q make unanimous peer pressure mathematically rare and scarce
- Boosted flipping probability $p_{inc} \Rightarrow$ ranges between 0.1 and 0.6 will effectively test whether mild versus highly aggressive retargeting campaigns can overcome induced churn

5.3 Reliability and repetitions

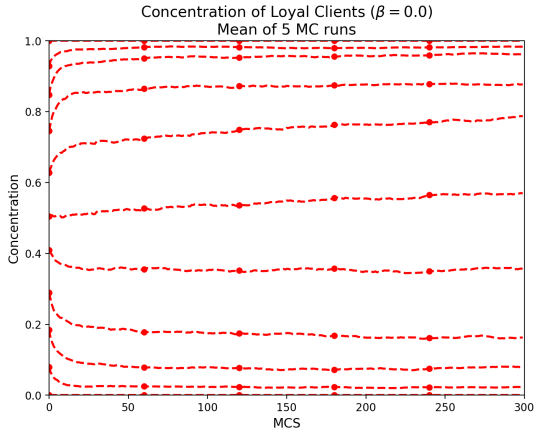
Because Monte Carlo simulations rely on pseudo-random individual agent interactions, single runs can be highly noisy. For general trends (showcased in the second tab), a lower number of repetitions (e.g., $num_mc_runs = 5$) provides a decent visual approximation of the mean trajectories while keeping computational time reasonable. As for statistical reliability (showcased in the first tab), to construct a robust 95% confidence interval and smooth out the inherent randomness, performing upwards of 200 repetitions would be advisable to achieve statistical certainty. The number of nodes is also important. A safe bet for statistically valuable results would be an upwards of 40000 nodes.

5.4 Quantitative and qualitative effects of the parameters

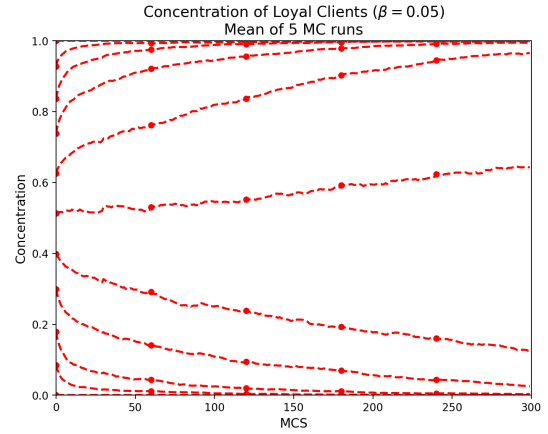
The parameters dictate the everchanging status between internal opinions, peer pressure, and marketing stimuli. One of them being peer group size of q , where a qualitative shift occurs as q increases. Because the q -voter model requires unanimous agreement among the q neighbors to force conformity, larger q values make it incredibly difficult for peers to override an agent's opinion. That's why the system transitions from a highly conformist society to one driven primarily by individual probabilities of initial churn rate. The second one is the network rewiring probability β . As it increases, random shortcuts are introduced throughout the graph, quantitatively decreasing the time it takes for an opinion or an ad to cascade globally. Finally the introduced mechanics of ads and their effectiveness which acts as system's recovery mechanism. A high p_{inc} qualitatively alters the system by acting as a shock absorber - even if a massive churn cascade begins, periodic ads will consistently rescue the network, preventing the system from ever reaching a 100% churn state.

5.5 Non-intuitive behaviors and explanations

When running these simulations, a certain non-linear or counter-intuitive phenomena can be observed, such as the small world non-linear shift, where changing β value from 0.0 to 0.05 causes a massive, disproportionate change in the churn rate (even though the change itself is minor), whereas moving it from 0.5 to 1.0 shows almost no difference. In Tab 2 plots we can also see that if the initial loyal concentration is 40%, the system entirely collapses to 0% loyalty. However, if the initial concentration is just slightly higher at 45%, it might soar to 100% loyalty.



(a) Concentration of loyal clients for $\beta = 0.0$



(b) Concentration of loyal clients for $\beta = 0.05$

6 GitHub and Streamlit links

- **Streamlit** =>
<https://q-voter-app-app-54xgh5gxxrr4pukoxwfmwe.streamlit.app/>
- **GitHub** =>
<https://github.com/hubertnowicki00/q-voter-streamlit-app>

7 Experimenting with the simulation's parameters

7.1 First experiment - the small world tipping point

The objective of this experiment was to precisely identify threshold of network randomness, where customer opinions shift from isolated local clusters to global interactions. To properly assess said thing we need the following parameters values:

- $N = 40000$
- $k = 8$
- $q = 4$
- $p_{inc} = 0.4$
- $initial_churn = 0.5$
- $\beta \in [0.0, 0.01, 0.02, 0.05, 0.1, 0.5]$
- Number of Monte Carlo simulations = 200
- Runs per concentration line = 10

With these values we could observe through plots how the mean churn rate changes between β values (0.5 being the end value, because as shown in the sensitivity analysis bigger β values make no significant changes to the simulation's output).

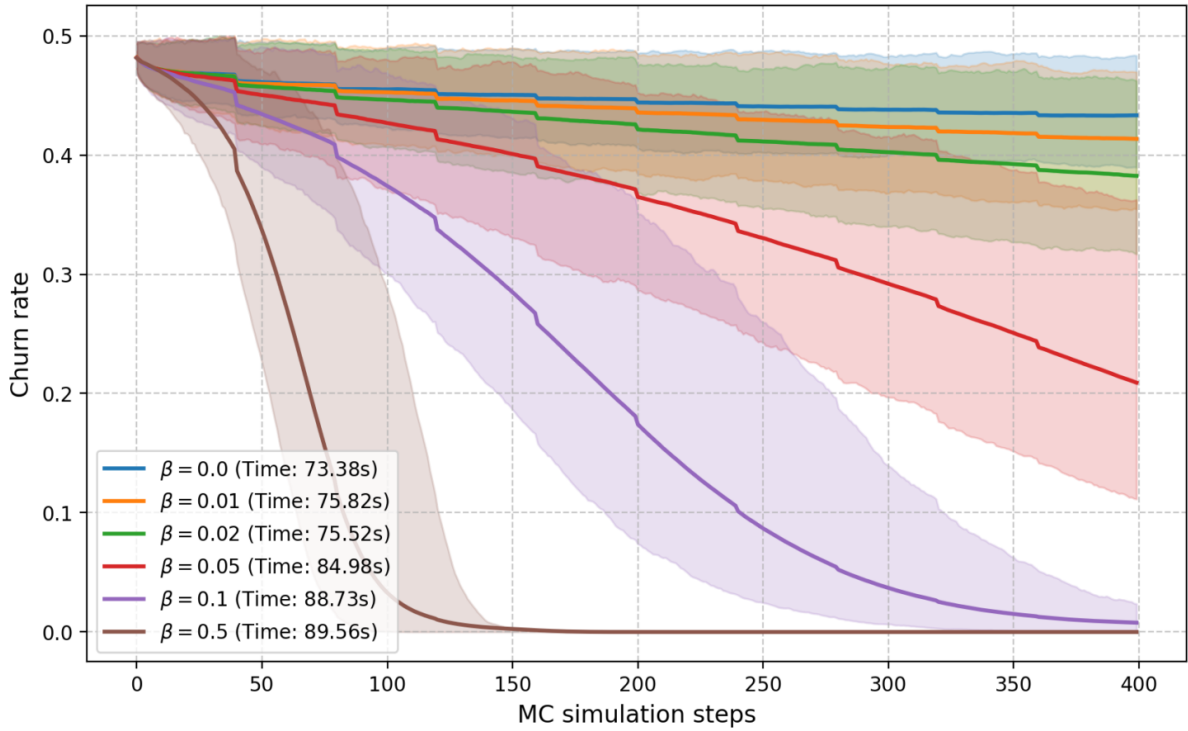
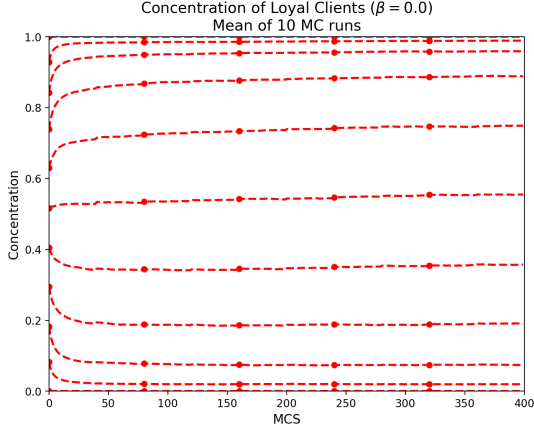
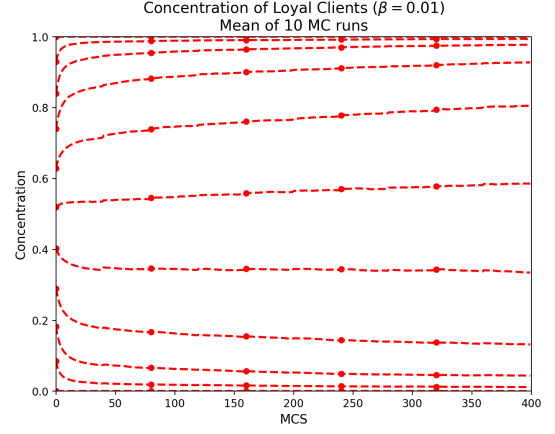


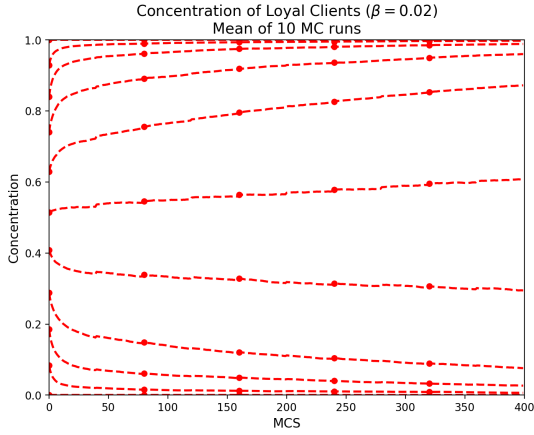
Figure 4: First experiment's customer churn plot



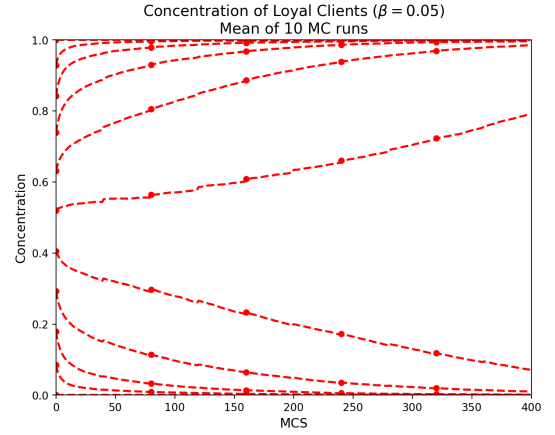
(a) Loyal clients concentration - $\beta = 0.0$



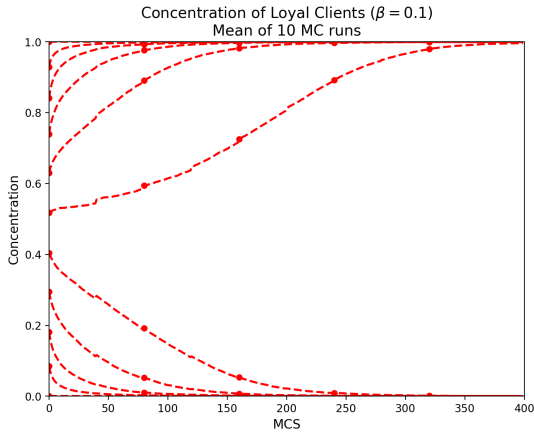
(b) Loyal clients concentration - $\beta = 0.01$



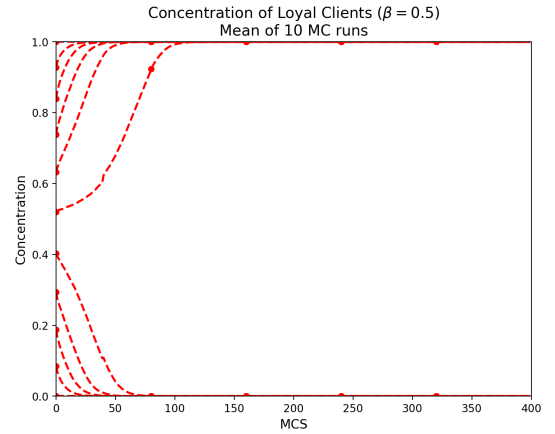
(a) Loyal clients concentration - $\beta = 0.02$



(b) Loyal clients concentration - $\beta = 0.05$



(a) Loyal clients concentration - $\beta = 0.1$



(b) Loyal clients concentration - $\beta = 0.5$

As the plot from the first tab shows, increasing the β value leads to a lower final churn rate. Since $\beta = 0$ represents a regular lattice and higher β values approach a random network, we can definitely see that the "small-world" effect efficiently facilitates better customer retention while episodic advertising by the company is applied. Moreover, we can also see that higher β values reach consensus faster than the other. Populations with

lower β values tend to reach equilibrium much longer. Finally, while generally the model's behavior for different rewiring probability does not change gradually, we could adopt a thesis of $\beta = 0.5$ being the tipping point, because of the sheer curve of it's population line on the plot.

7.2 Second experiment - the conformity barrier

The objective of the second experiment was to understand how the size of a customer's group represented by q suppresses or accelerates the effectiveness of advertising. Out of care for the significance of said experiment's results we need the following parameters values:

- $N = 40000$
- $k = 8$
- $initial_churn = 0.3$
- Rewiring probability $\beta = 0.2$
- $p_{inc} = 0.3$ (moderate ad effectiveness)
- $q \in [2, 3, 4, 5]$
- Number of Monte Carlo simulations = 200
- Runs per concentration line = 10

7.2.1 Results for $q = 2$

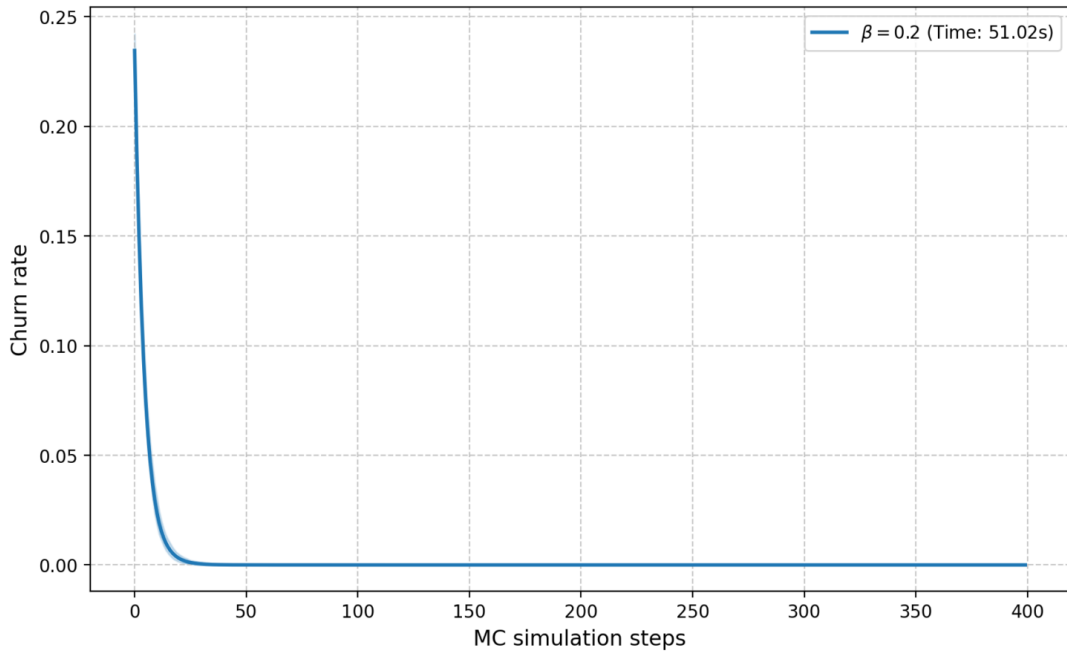


Figure 8: Second experiment's customer churn plot for $q = 2$

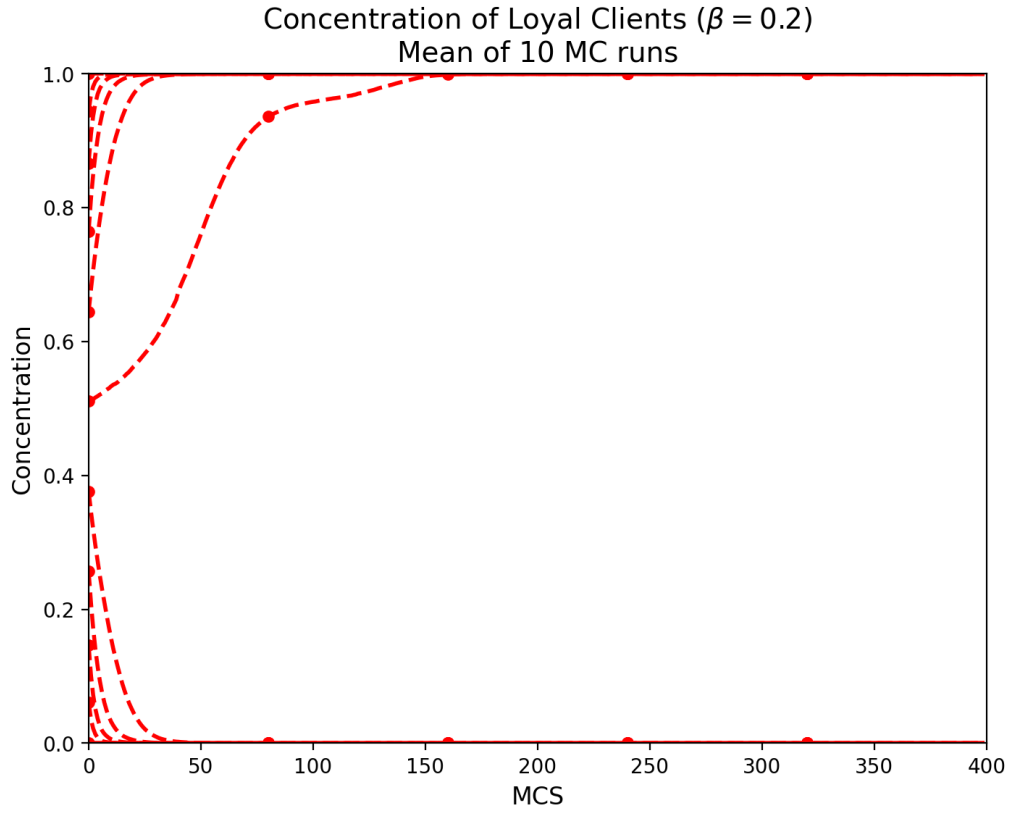


Figure 9: Second experiment's concentration of loyal customers plot for $q = 2$

7.2.2 Results for $q = 3$

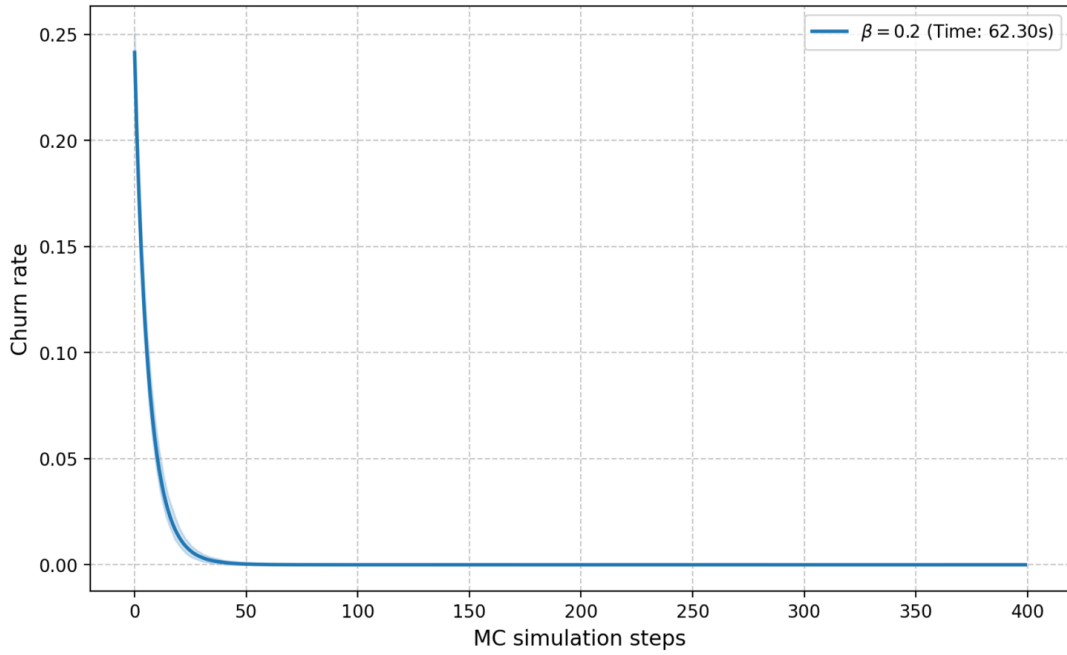


Figure 10: Second experiment's customer churn plot for $q = 3$

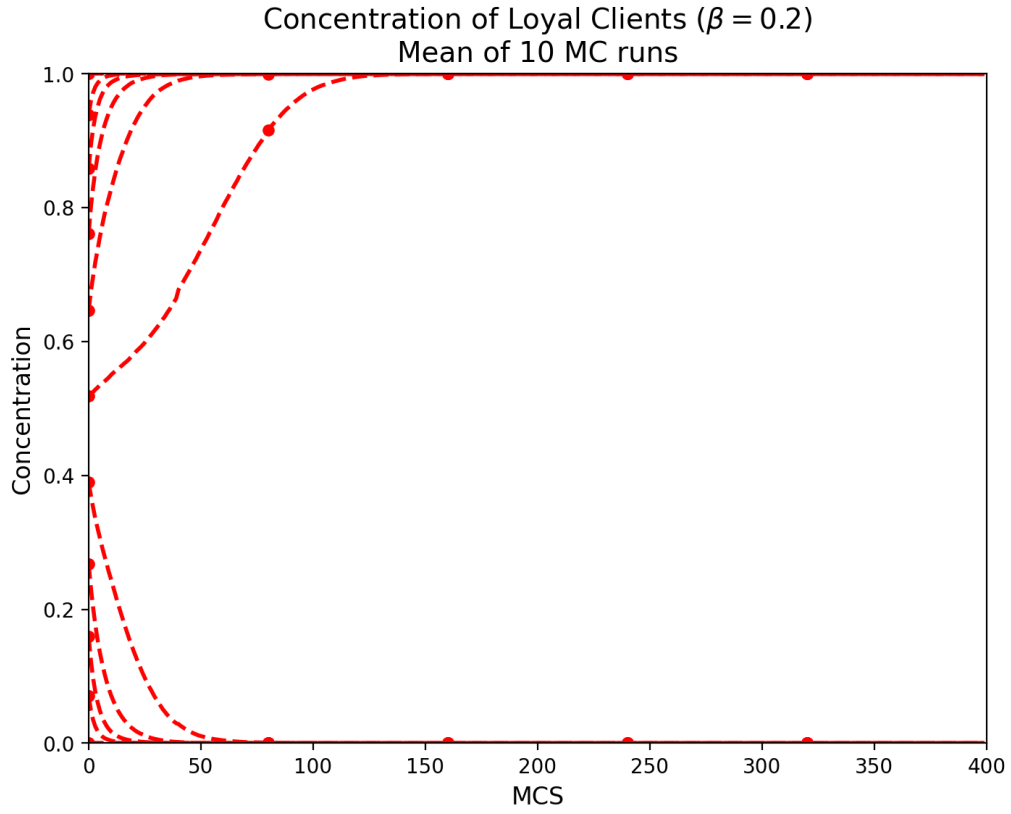


Figure 11: Second experiment's concentration of loyal customers plot for $q = 3$

7.2.3 Results for $q = 4$

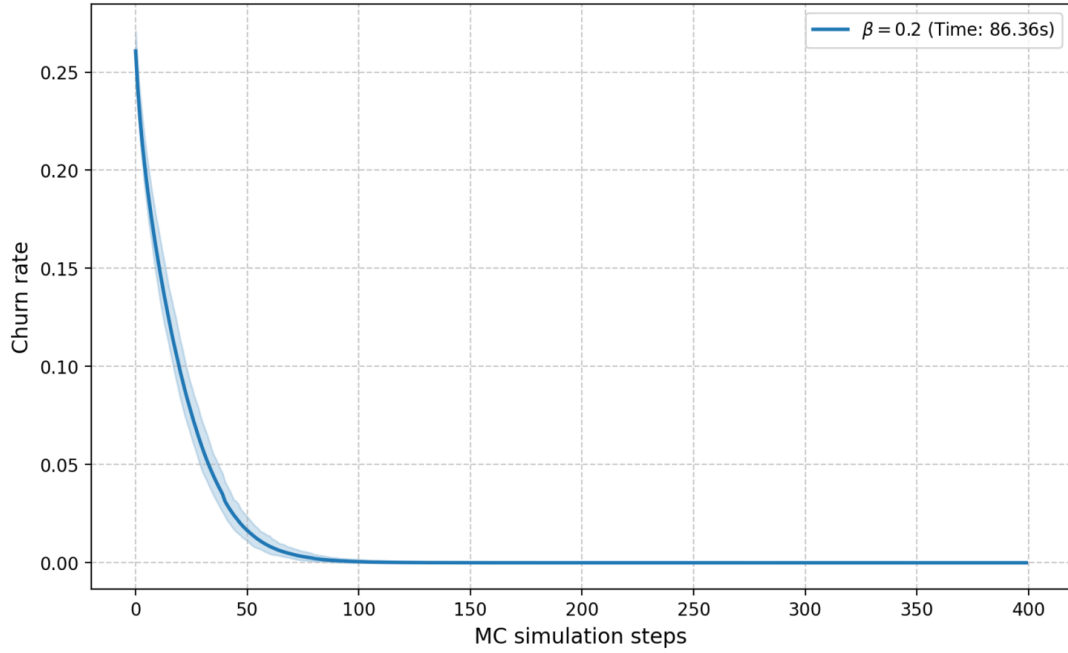


Figure 12: Second experiment's customer churn plot for $q = 4$

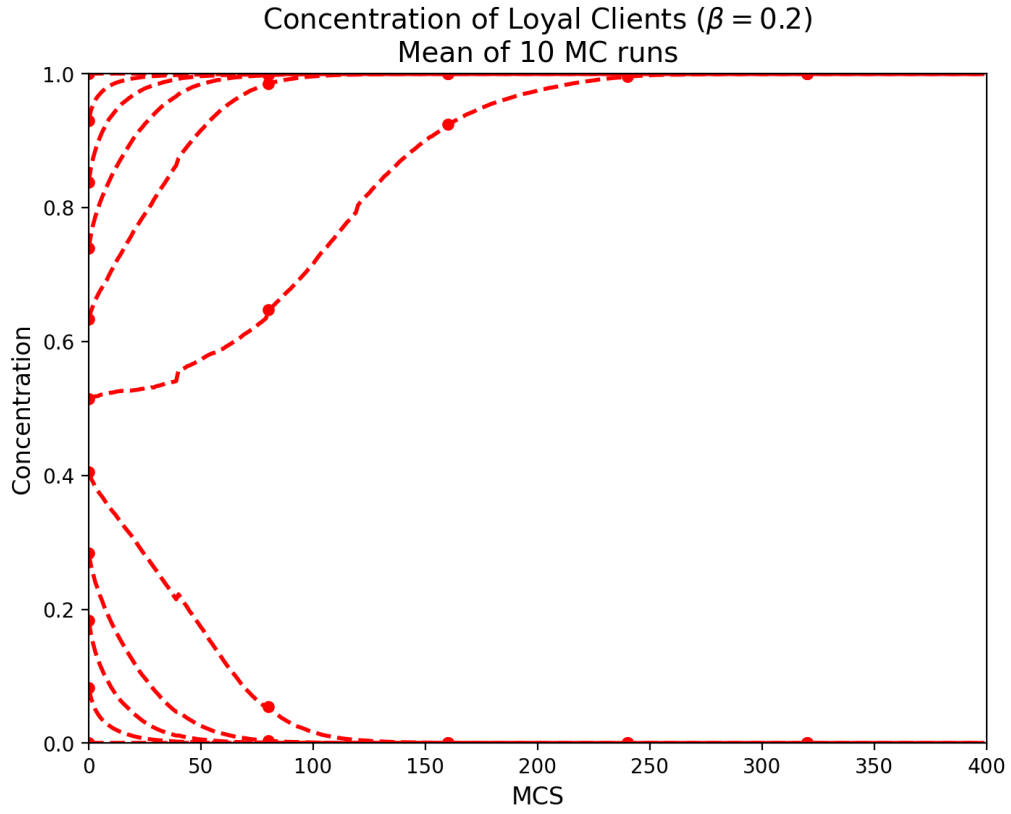


Figure 13: Second experiment's concentration of loyal customers plot for $q = 4$

7.2.4 Results for $q = 5$

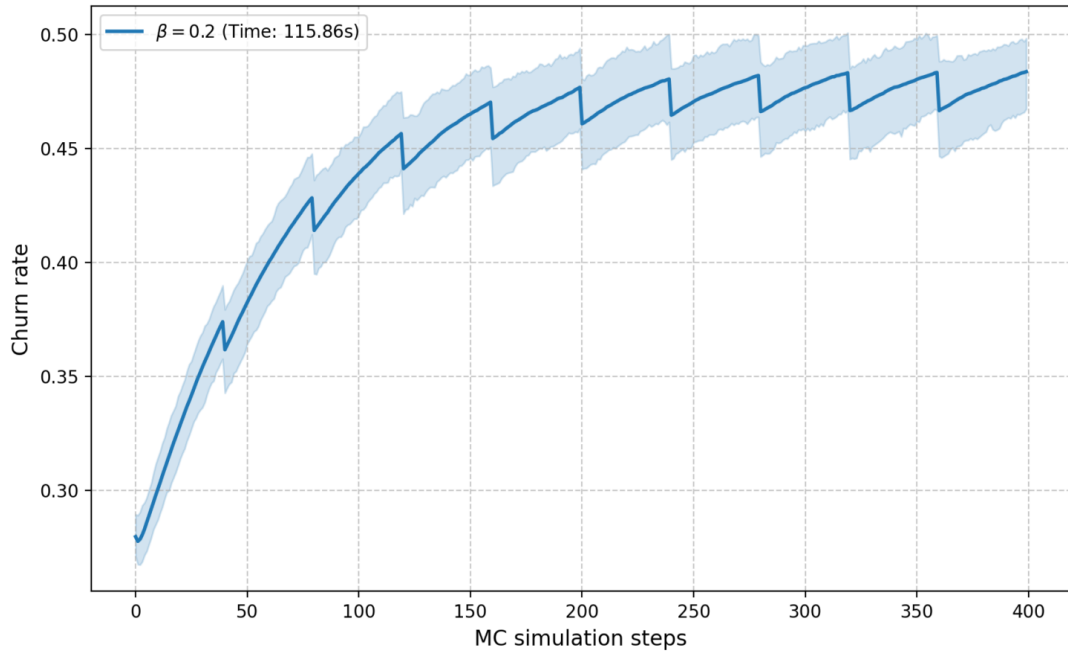


Figure 14: Second experiment's customer churn plot for $q = 5$

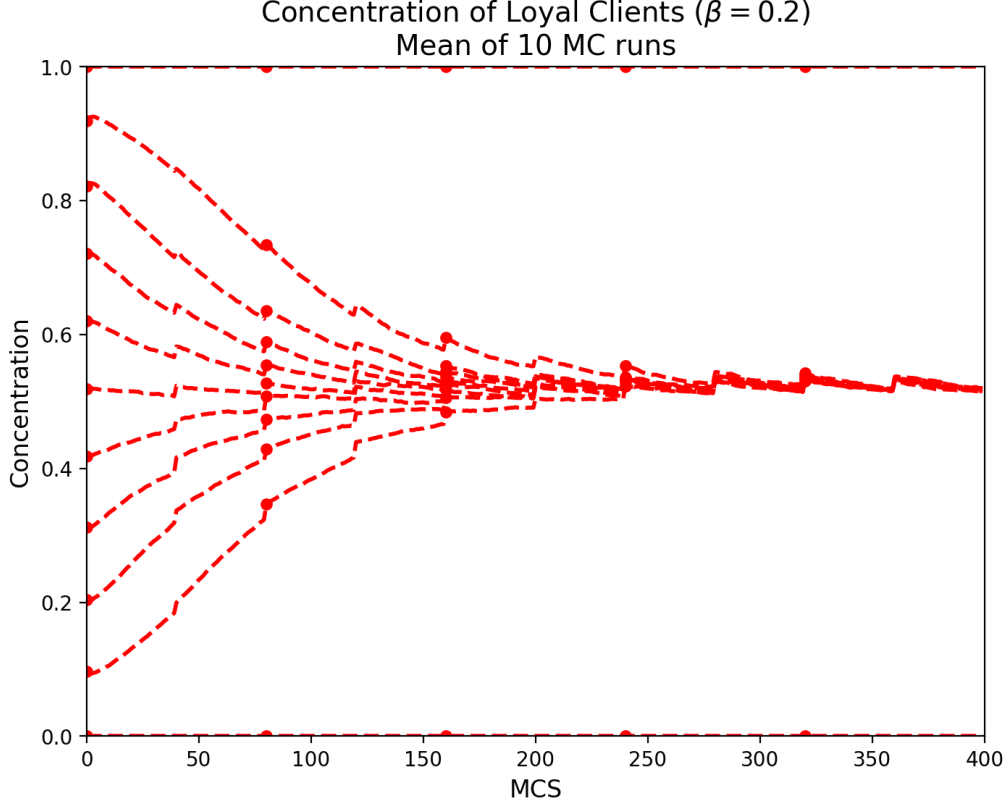


Figure 15: Second experiment’s concentration of loyal customers plot for $q = 5$

7.2.5 Results overview

The empirical trajectories of above plots reveal a distinct threshold in the system’s susceptibility to social reinforcement. While influence groups of $q = 2, 3$, and 4 maintained relative stability and similar behavior (consequently lowering the churn rate), expanding the required consensus to $q = 5$ induced a drastic phase shift that increased the churn rate. This indicates that excessively high conformity barriers actively suppress the propagation of advertising loyalty, as the probabilistic likelihood of an agent selecting a unanimous group of neighbors influenced by said ads diminishes significantly. Furthermore, a quick analysis of loyal client concentration plots demonstrates that only one of the groups (for $q = 5$) tends to reach an equilibrium of 0.5 , being in complete opposition to the polarized rest.

These phenomena have an underlying mathematical explanation. Since agents require unanimous agreement among q neighbors to adopt their state without referencing their own flipping probability, requiring 5 neighbors to all be identical is exponentially harder than requiring 2 or 3 . At $q = 5$, the conformity rule almost never triggers, meaning agents rely almost entirely on their default, un-boosted baseline flipping probabilities, allowing natural churn to tear through the network.

7.3 Third experiment - marketing ROI

The objective of the third experiment was to find the minimum boosted flip probability required to reverse a dying customer base and return the system to profitability. Let's assess a harsh scenario of:

- $initial_churn = 0.8$
- $\beta = 0.2$
- $q = 4$
- $p_inc \in [0.1, 0.2, 0.3, 0.4, 0.5]$

with the rest of parameters the same as before.

7.3.1 Results for $p_inc = 0.1$

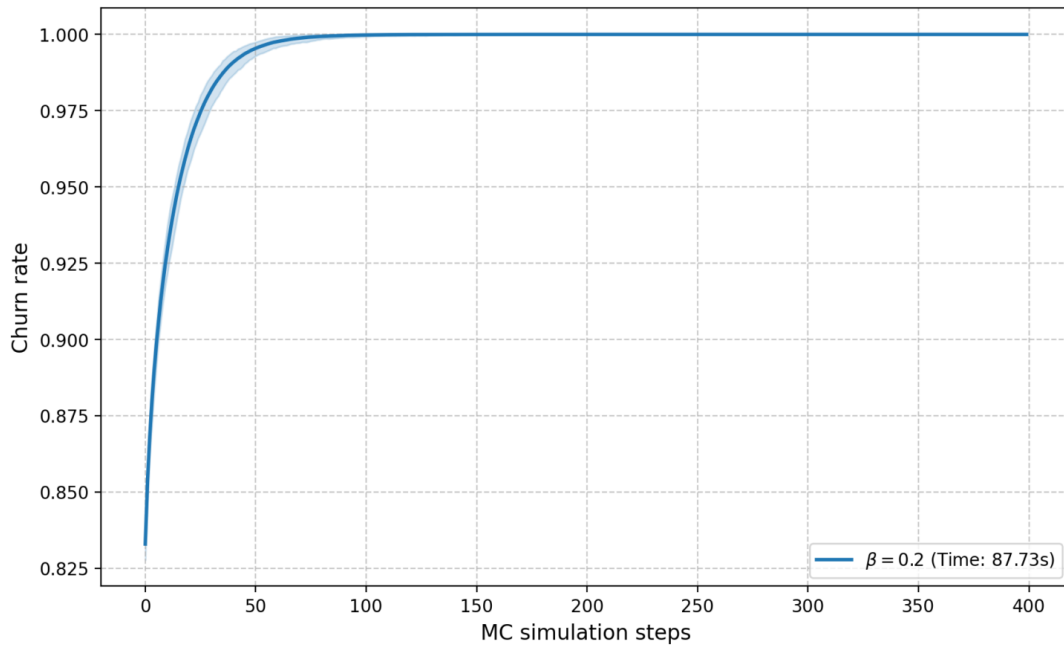


Figure 16: Third experiment's customer churn plot for $p_inc = 0.1$

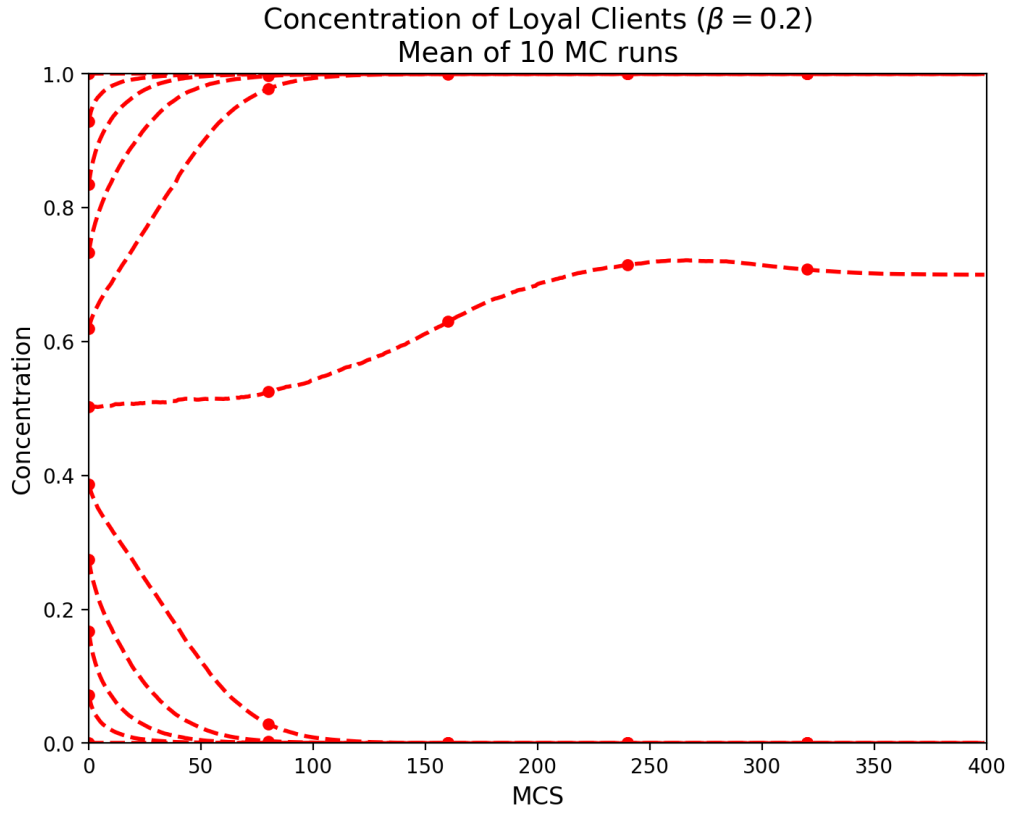


Figure 17: Third experiment's concentration of loyal customers plot for $p_{inc} = 0.1$

7.3.2 Results for $p_{inc} = 0.2$

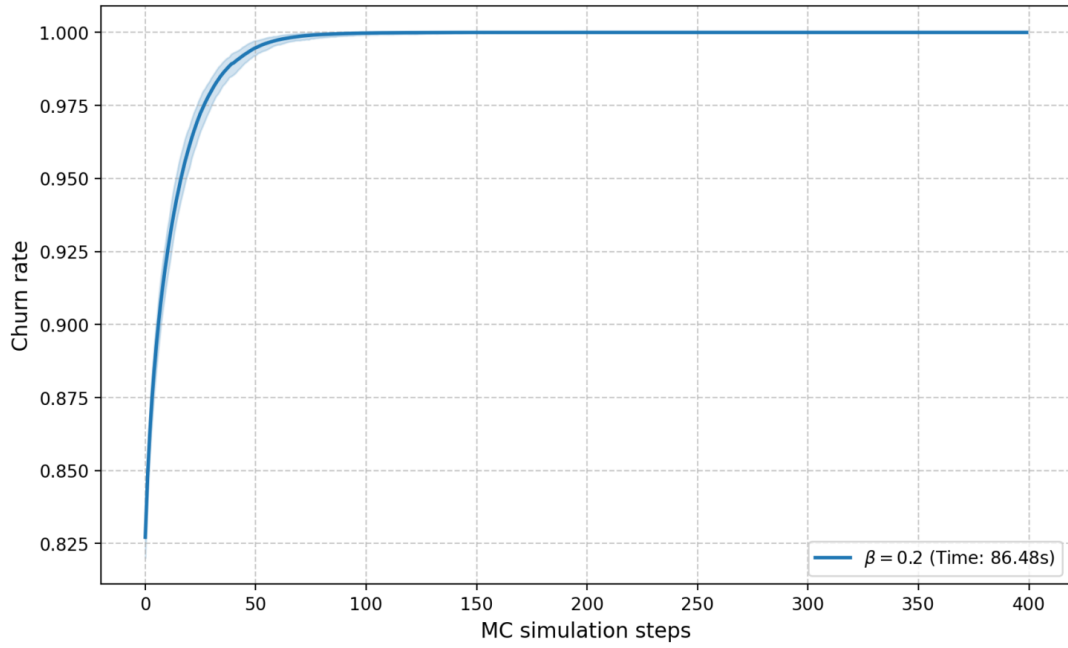


Figure 18: Third experiment's customer churn plot for $p_{inc} = 0.2$

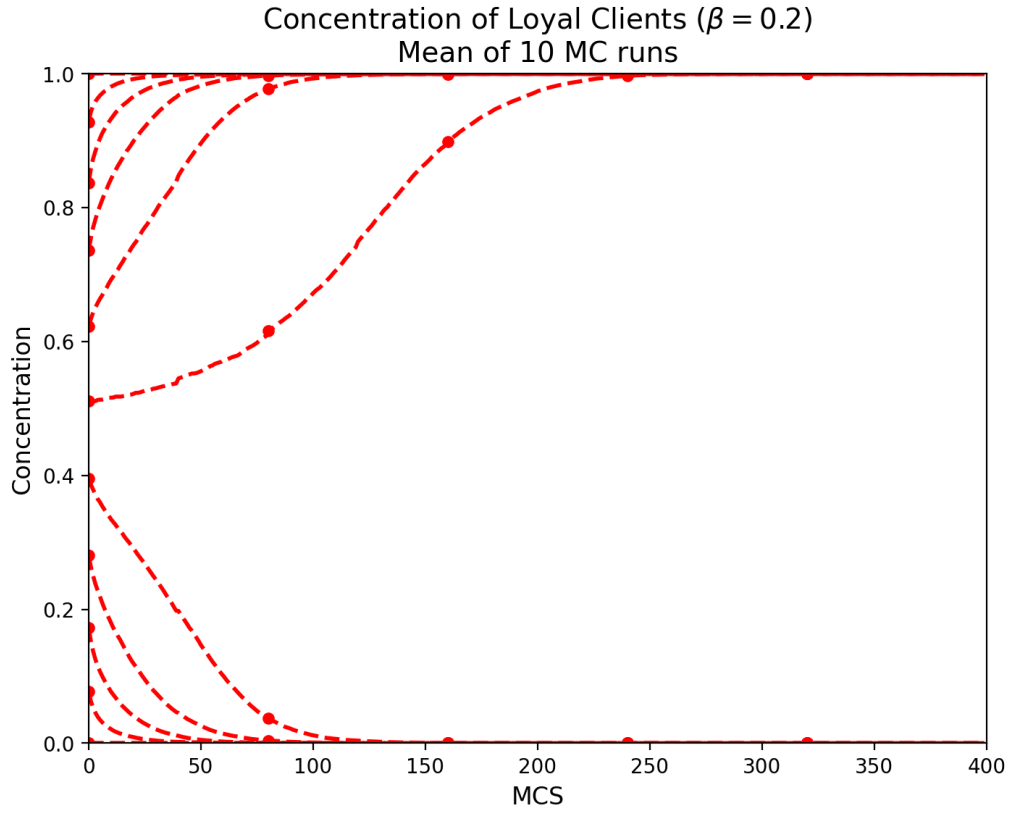


Figure 19: Third experiment's concentration of loyal customers plot for $p_{inc} = 0.2$

7.3.3 Results for $p_{inc} = 0.3$

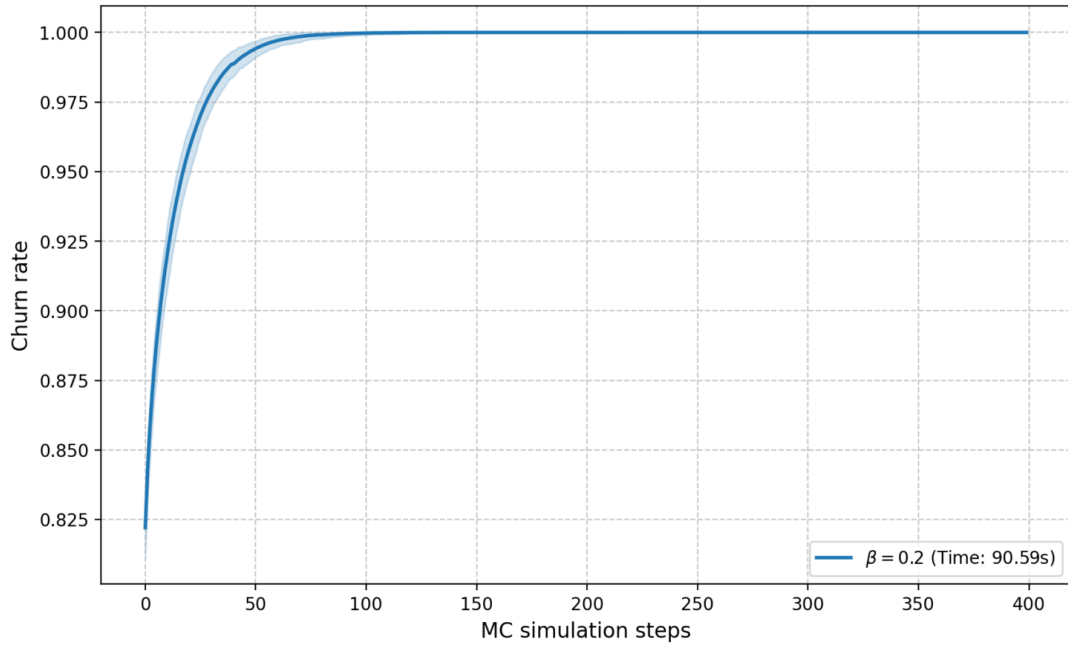


Figure 20: Third experiment's customer churn plot for $p_{inc} = 0.3$

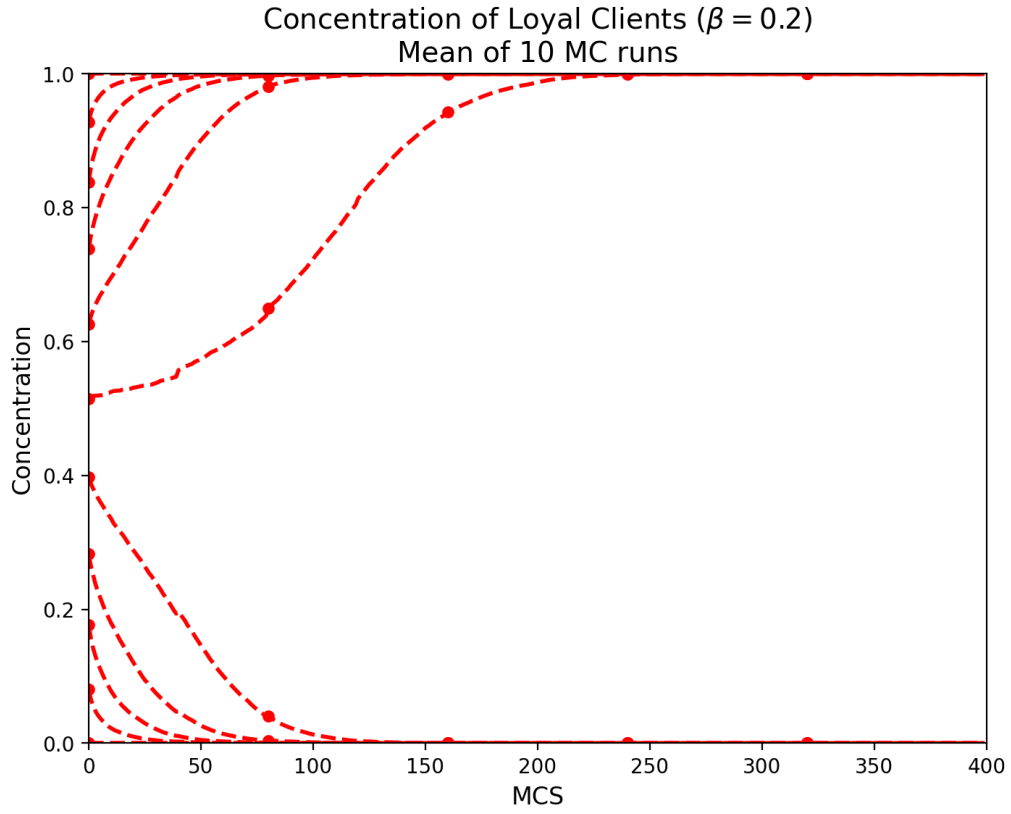


Figure 21: Third experiment's concentration of loyal customers plot for $p_{inc} = 0.3$

7.3.4 Results for $p_{inc} = 0.4$

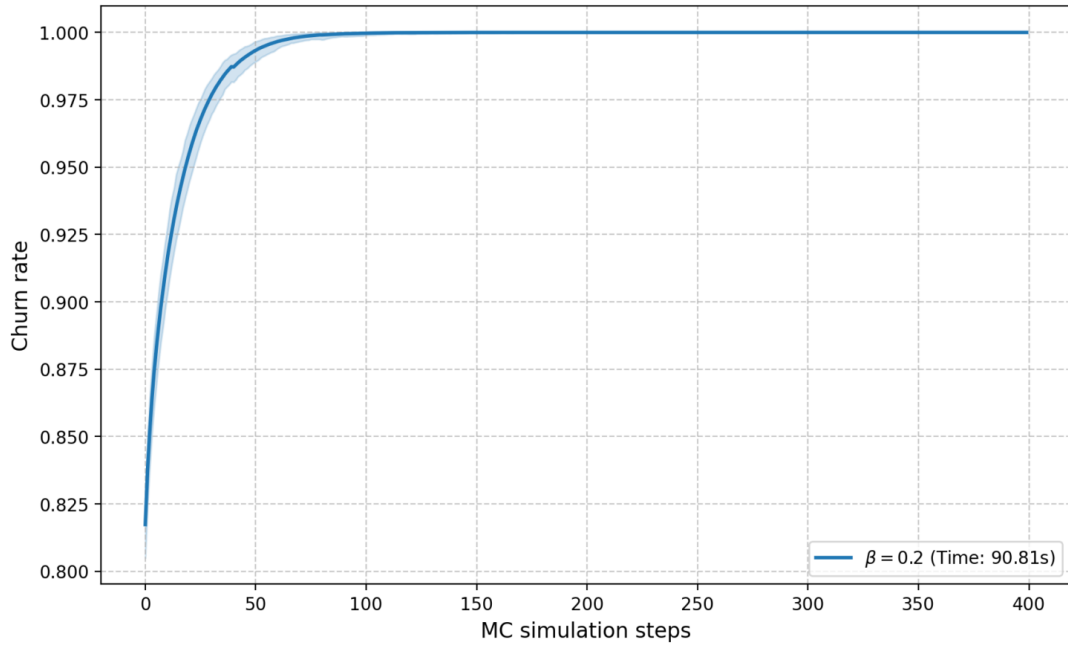


Figure 22: Third experiment's customer churn plot for $p_{inc} = 0.4$

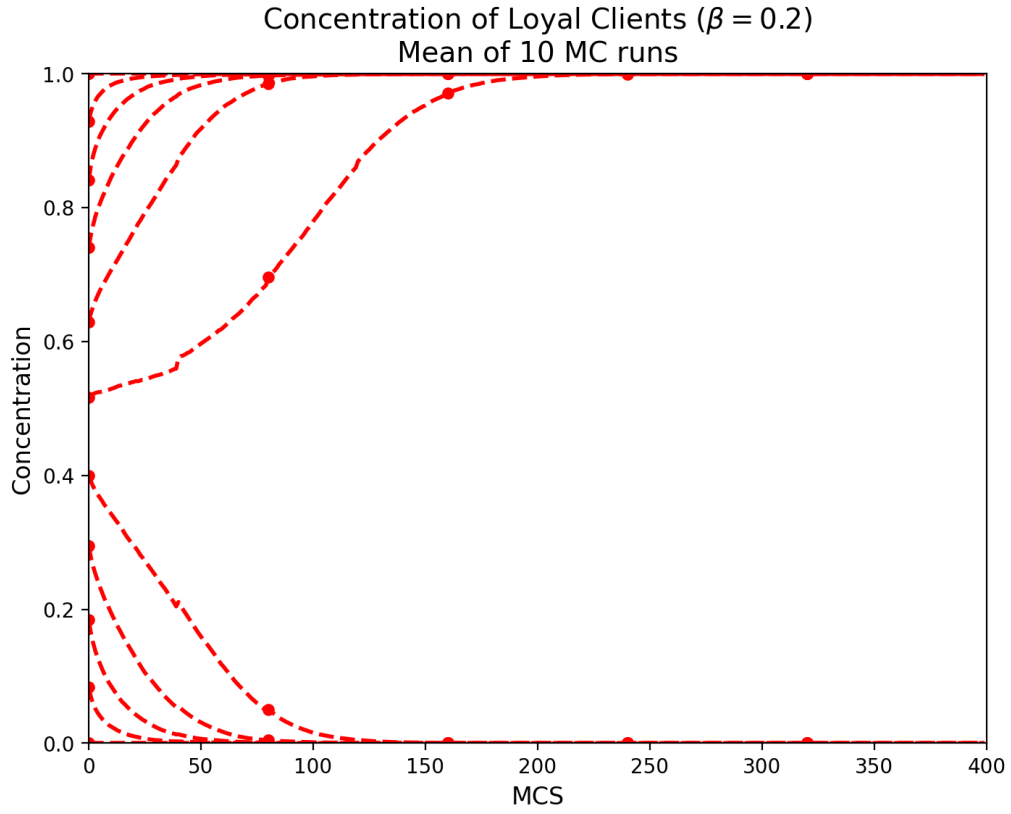


Figure 23: Third experiment's concentration of loyal customers plot for $p_{inc} = 0.4$

7.3.5 Results for $p_{inc} = 0.5$

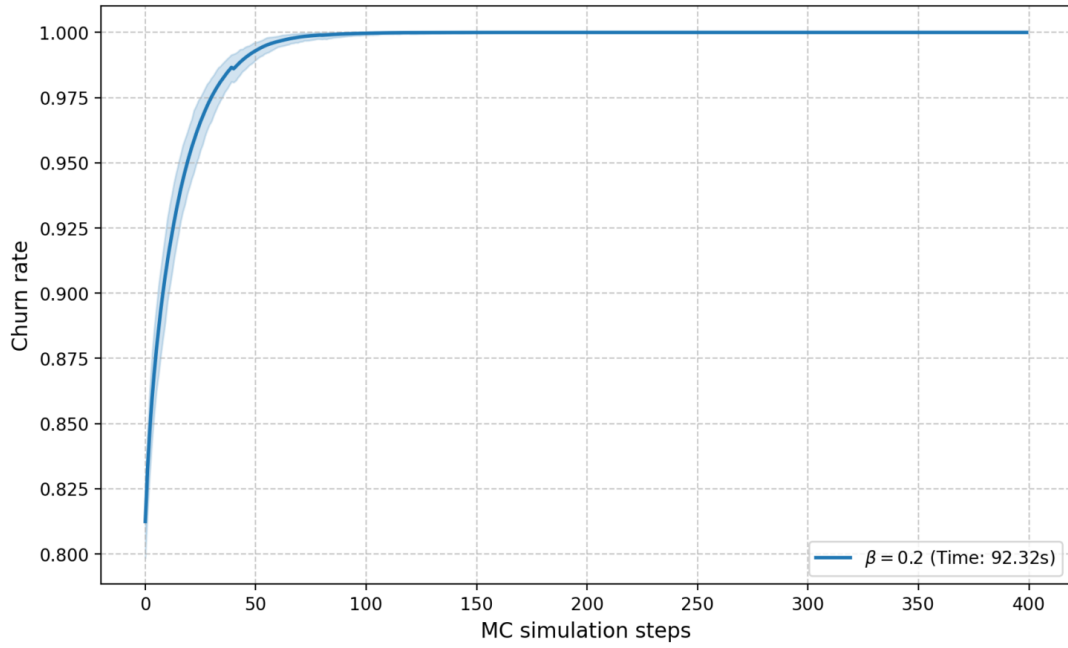


Figure 24: Third experiment's customer churn plot for $p_{inc} = 0.5$

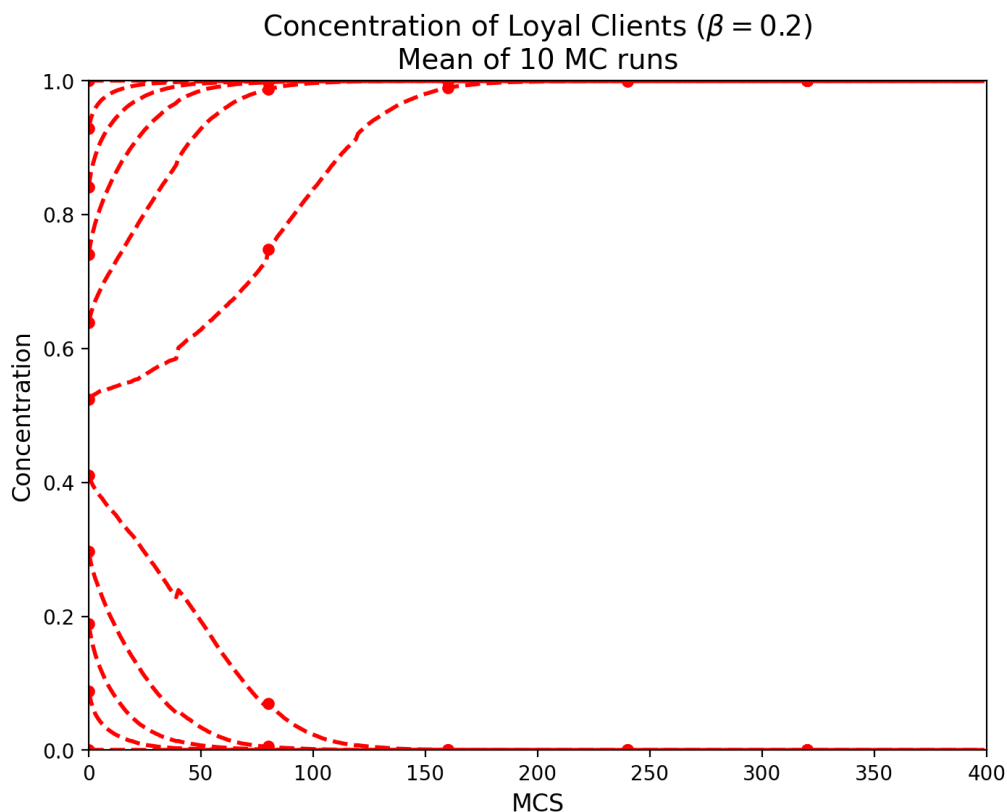


Figure 25: Third experiment's concentration of loyal customers plot for $p_{inc} = 0.5$

7.3.6 Results overview

The visual data conclusively demonstrates that under the specified structural conditions, none of the tested episodic marketing interventions were capable of halting the system’s collapse. Across all five efficacy scenarios, the macroscopic customer churn trajectories rapidly escalate from their initial deficit and unconditionally converge toward an absorbing state of 1.0, representing absolute customer loss. Furthermore, a cross-sectional analysis of the loyal client concentration mappings reveals that while a higher p_{inc} does marginally lower the system’s structural tipping point, the 0.2 initial loyal fraction remains irreparably trapped within the negative basin of attraction across the entire parameter range. Ultimately, these empirical results establish a harsh boundary condition for marketing ROI, that if once a complex small-world network reaches a critical density of churned agents, the localized unanimous peer pressure inherently overpowers the asymmetric benefits of retargeting campaigns, rendering standard marketing efforts entirely futile.

7.4 Fourth experiment - bistability and attractor mapping

The fourth experiment's objective was mapping the dead ends in the customer base by identifying the system's natural attractors. To achieve statistically significant results the following values of parameters had been adopted:

- $\beta = 0.05$
- $q = 4$
- $p_{inc} = 0.15$ (weak ads to let pressure dominate)
- $num_mc_runs = 10$
- $initial_churn = 0.5$

with the rest of parameters the same as before.

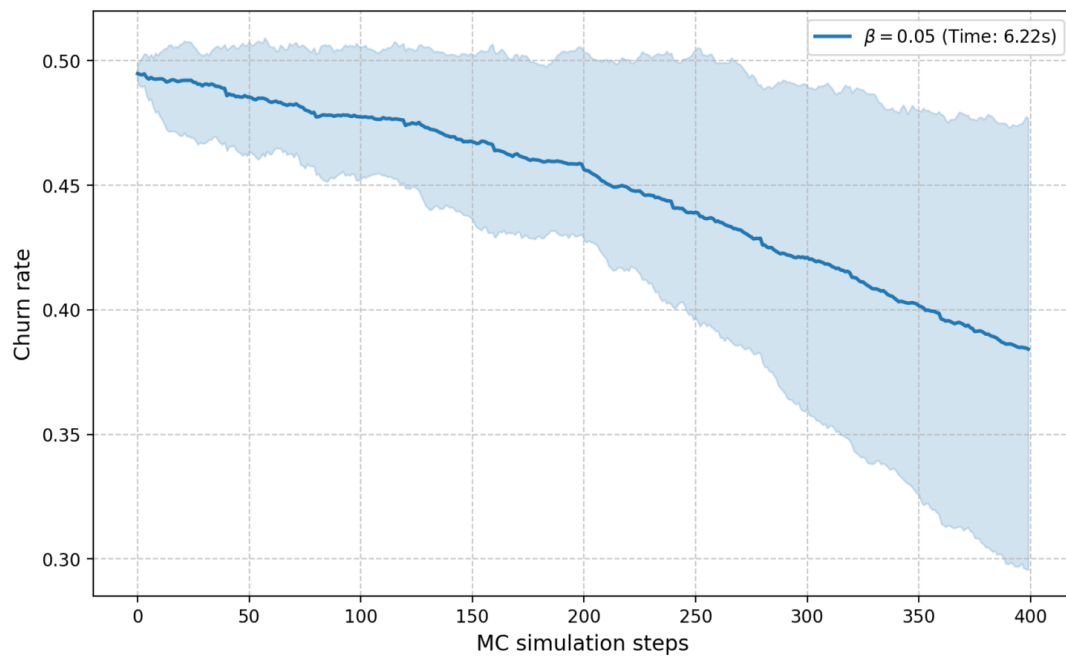


Figure 26: Fourth experiment's customer churn plot

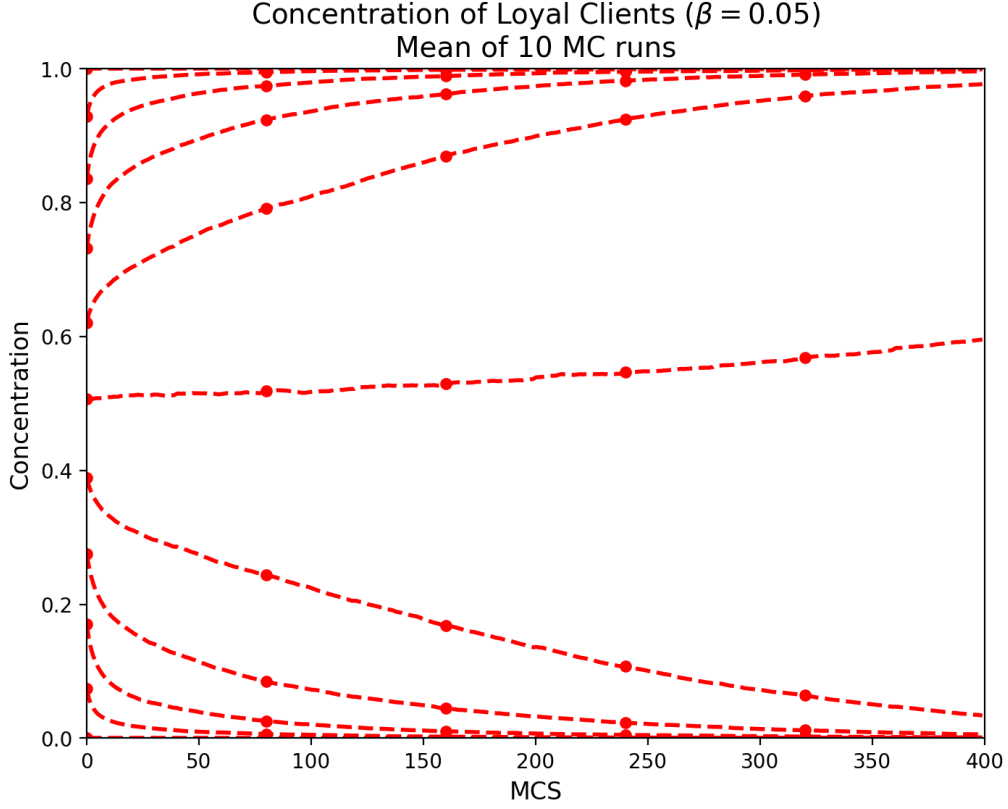


Figure 27: Fourth experiment's concentration of loyal customers plot

While looking at the concentration of loyal clients plot, we can clearly see that the phase space split. The trajectories starting at 0.4 and below swiftly cascade down to the total churn, while the ones starting at 0.6 and up swiftly cascade upwards. The only exception is the middle line starting at 0.5 - as expected, it doesn't immediately collapse or jump. Instead, it hovers in a metastable state and very slowly creeps upwards. This is the perfect visualization of the slight symmetry-breaking effect of a weak episodic ads pushing back against the negative pressure of the established customer base. Combined with the churn rate plot, which clearly shows an extreme variance we can deduce that in this specific model only two attractors are present - 0 and 1. And they are responsible for such highly polarized results in any scenario.

8 Conclusion for the entire project

In conclusion, this project successfully demonstrates that customer churn is fundamentally a non-linear, structurally driven process rather than a mere aggregation of isolated individual decisions. By integrating a modified q-voter model with episodic marketing stimuli across a Watts-Strogatz graph, the simulations reveal that the efficacy of advertising is strictly bounded by the network’s structural parameters, specifically the conformity barrier q and the degree of rewiring probability β . Crucially, the phase space mappings confirm a system characterized by aggressive bistability, wherein the customer base inevitably cascades toward total loyalty or absolute macroscopic churn depending on its position relative to a highly sensitive separatrix. The empirical findings — most notably the inability of moderate to aggressive marketing to reverse a deeply entrenched churn cascade — underscore a vital strategic imperative. The intervention must be preemptive.